EEG-Based Analysis of Alzheimer's Disease

Exploratory Data Analysis Report Individual Datasets and Merged Dataset

1. Exploratory Analysis Overview

This report presents the results of an exploratory analysis of EEG (brainwave) recordings from people with Alzheimer's disease (AD) and healthy individuals. The aim is to understand the data, identify its limitations, find the most important

2. Data Sources

This analysis is based on two publicly documented EEG datasets, both associated with prior research on quantile graphs and computational EEG methods for Alzheimer's disease classification. Details and access information for each are listed below.

Dataset 1

Campanharo, A. S. L. O., Ramos, F. M., Pineda, A. M., and Betting, L. E. "Data from: Quantile Graphs for EEG-Based Diagnosis of Alzheimer's Disease," 2020.

DOI: 10.17605/OSF.IO/S74QF

Online: https://osf.io/pa38y

Dataset 2

Vicchietti, M. L., Ramos, F. M., Betting, L. E., and Campanharo, A. S. L. O. "Data from: Computational Methods of EEG Signals Analysis for Alzheimer's Disease Classification," 2023.

DOI: 10.17605/OSF.IO/2V5MD

Online: https://osf.io/2v5md/

3. How Data Looks

The EEG recordings were divided into five standard frequency bands: delta, theta, alpha, beta, and gamma. These bands were analyzed across the whole brain and in different brain regions (frontal, central, temporal, parietal, and occipital).

The analysis showed several clear patterns:

- Delta had the highest power in both datasets, accounting for about 33% of total power in Dataset 1 and 37% in Dataset 2.
- Theta power was generally higher in people with Alzheimer's disease, especially in the parietal and central regions.
- Healthy participants showed stronger alpha activity, particularly in the occipital region (O1 and O2), which is a normal pattern in healthy brains.
- Beta and gamma activity were similar in both groups, with only small differences.

- t-SNE visualization showed that Dataset 2 had a clearer separation between Healthy and AD recordings, while Dataset 1 showed more overlap between the two groups.

In short, both datasets tell a similar story — AD brains show a shift toward slower brain activity (more theta, less alpha) but Dataset 2 shows this pattern more cleanly than Dataset 1.

4. Gaps and Limitations Found in the Data

While reviewing the data, a few gaps and weaknesses stood out that should be kept in mind before building any classification model:

- Small sample sizes: Dataset 2 has only 24 people per group, and Dataset 1 has an uneven split (12 healthy vs. 80 AD). Small and unequal group sizes make results less reliable and easier to over-interpret.
- Class imbalance in Dataset 1: with far more AD samples than healthy samples, any model trained on this data alone could become biased toward predicting AD too often.
- Unstable grouping in Dataset 1: the t-SNE visual grouping changes depending on settings and does not clearly separate the two groups, meaning the raw feature space in this dataset may be noisier or less distinct.
- Redundant features: some features carry almost the same information twice, such as the theta/alpha ratio and the alpha/theta ratio, which are just each other's mirror image. Using both in a model does not add new information and can distort results.
- Single-recording comparisons: some of the frequency spectrum comparisons (FFT plots) were based on just one recording from each group. These are useful for a first look but are not something we can generalize from — they need to be confirmed at the group level.
- Borderline features: Hjorth mobility, a signal-complexity measure, was almost statistically significant in one dataset (just barely missing the cutoff after correcting for multiple comparisons). It may still hold useful information but cannot be relied on as-is.

None of these gaps mean the data is unusable hey simply mean results should be treated as an early-stage, exploratory finding rather than something ready for clinical use, and should be validated on a larger, more balanced dataset before any real-world application.

5. Which Features Matter Most

After testing many EEG features across both datasets, a small group of features consistently stood out as the strongest at telling AD and healthy participants apart. These are summarized below.

Feature	Pattern in AD	Why it matters
Parietal Theta	Much higher in AD	The single strongest and most consistent marker across every analysis in this report
Relative Theta (overall)	Higher in AD	Very large, consistent effect size in both datasets; easy to measure
Temporal / Occipital / Central Theta	Higher in AD	Strong and reliable, just slightly behind Parietal Theta
Central Delta	Lower in AD	One of the strongest "opposite direction" markers — healthy people show more of it

Feature	Pattern in AD	Why it matters
Theta/Alpha Ratio	Higher in AD	Captures the shift from fast to slow brain activity in one number
Relative Delta	Lower in AD	Reliable, moderate-strength marker
Alpha Peak Frequency	Lower in AD	Reflects a slower dominant brain rhythm in AD
Hjorth Activity & Complexity	Lower in AD	Signal-complexity measures — AD brain signals appear "simpler" or less variable
Spectral Entropy	Higher in AD	Measures how spread out the signal's energy is across frequencies
Frontal Theta / Delta	Weaker but still useful	Still significant, but less powerful than parietal or central regions

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Some EEG features did not show clear or consistent differences between people with Alzheimer's disease (AD) and healthy participants. These include relative alpha, relative beta, relative gamma, the delta/alpha ratio, and Hjorth mobility (in one dataset). Because these features do not reliably distinguish the two groups, they are not good choices to use as individual features in a classification model.

One important point is that the theta/alpha ratio and the alpha/theta ratio contain the same information because one is simply the inverse of the other. Therefore, only one of these ratios should be included in a classification model. Using both does not improve performance and only makes the model more complex.

6. How This Data Could Be Used for Classification

Based on the findings of this exploratory analysis, the next step is to develop a classification model that can distinguish people with Alzheimer's disease (AD) from healthy individuals. A practical approach is to use the most informative EEG features, such as parietal theta power, relative theta, theta/alpha ratio, central delta, alpha peak frequency, Hjorth parameters, and spectral entropy, while excluding features that showed little or no difference between the two groups. Focusing on these reliable features can improve classification performance, reduce unnecessary complexity, and provide a strong foundation for building an accurate and efficient AD detection model.

6.1 Feature Selection

- Start with the features that showed the largest, most consistent effect sizes: Parietal Theta, overall Relative Theta, Central Delta, Theta/Alpha Ratio, Alpha Peak Frequency, Hjorth Activity, Hjorth Complexity, and Spectral Entropy.
- Drop features that were not statistically significant (relative alpha/beta/gamma, delta/alpha ratio) unless later testing shows they add value in combination with other features.
- Keep only one version of each ratio pair (e.g., keep Theta/Alpha, drop Alpha/Theta) to avoid redundancy.

7. Overall Conclusion

Across both individual datasets and the merged dataset, the same clear pattern shows up: Alzheimer's disease is associated with a slowing of brain activity more theta, less delta and alpha, lower signal complexity, and a slower dominant brain rhythm. Parietal Theta stands out as the single most reliable marker, supported by relative theta, central delta, and a handful of related features.

At the same time, the data has real limitations small and imbalanced samples, some unstable grouping in one dataset, and a few borderline or redundant features. These findings are a strong and promising starting point for building a classification model, but they should be treated as exploratory results that need validation on a larger dataset before being used for any real diagnostic purpose.